

Artificial Intelligence Lab

Lab#1

Course Name	Artificial Intelligence
Course Code	CSC 413
Semester	Fall 2026
Program	BS
Credits	1
Faculty	Madeha Arif

Python Fundamentals – Decisions, Match-Case and Lists

1. Objectives

By the end of this experiment, the student will be able to:

- Understand what Python is and where it is used
- Install and configure Python 3.12.1 on a Windows system
- Write and run simple Python programs
- Understand basic Python syntax: variables, indentation, comments, numbers, strings, lists
- Use simple built-in modules (math, random)
- Define and call simple functions
- Build small beginner-level console programs

2. Required Tools / Equipment

Item	Description
Operating System	Windows 10 / 11 (64-bit)
Software	Python 3.12.1
Editor	IDLE (comes with Python)
Terminal	Command Prompt (cmd)
Internet	Required only for downloading Python

3. Theory

3.1 Introduction to Python

Python is a popular programming language created by Guido van Rossum and released in 1991. The current version is Python 3.

It is simple, readable, and easy to learn — perfect for beginners.

3.2 Applications of Python

- Web Development (server-side)
- Software Development
- Mathematics
- System Scripting
- Artificial Intelligence and Data Science

3.3 What Can Python Do?

- Create web applications on a server
- Build workflows alongside other software
- Connect to databases and modify files
- Handle big data and perform complex mathematics
- Support rapid prototyping and production-ready development

3.4 Why Python?

- Works on Windows, Mac, Linux, Raspberry Pi
- Syntax is simple and similar to English
- Requires fewer lines of code than other languages
- Interpreted — code runs as soon as it is written
- Supports procedural, object-oriented, and functional styles

3.5 Python vs Other Languages

Feature	Python	C / Java
Line ending	New line	Semicolon ;
Block	Indentation	Curly braces { }
Readability	Very high	Moderate

3.6 Core Concepts

- Variables — boxes that store values: $x = 5$
- Indentation — spaces at the start of a line (usually 4 spaces) that define a block
- Comments — start with #, ignored by Python

- Numbers — + - * / work like a calculator
- Strings — text inside '...' or "..."
- Lists — many items inside [...]

4. Procedure

4.1 Installation of Python 3.12.1

- Open browser → visit python.org/downloads
- Download Python 3.12.1 (Windows 64-bit)
- Run the installer
- Check "Add Python 3.12 to PATH" at the bottom
- Click Install Now
- Open Command Prompt and type:

```
python --version
```

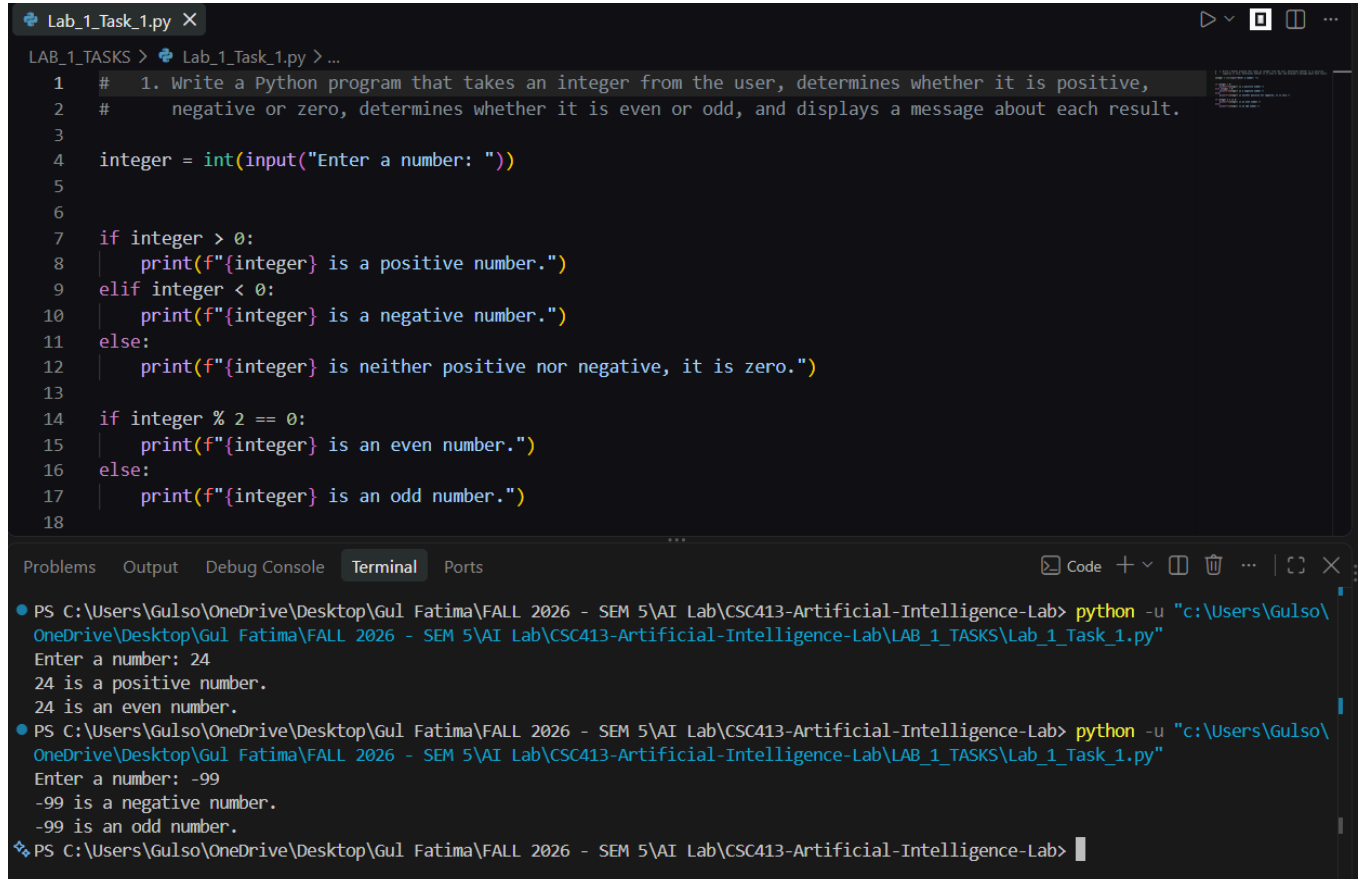
4.2 Writing and Running a Program

- Open IDLE (search "IDLE" in Start Menu)
- Click File → New File
- Write your code
- Save with .py extension (e.g., program1.py)
- Press F5 to run

Lab Task

1. Write a Python program that takes an integer from the user, determines whether it is positive, negative or zero, determines whether it is even or odd, and displays a message about each result.

Code & Output:



```
Lab_1_Task_1.py X
LAB_1_TASKS > Lab_1_Task_1.py > ...
1  # 1. Write a Python program that takes an integer from the user, determines whether it is positive,
2  #     negative or zero, determines whether it is even or odd, and displays a message about each result.
3
4  integer = int(input("Enter a number: "))
5
6
7  if integer > 0:
8      print(f"{integer} is a positive number.")
9  elif integer < 0:
10     print(f"{integer} is a negative number.")
11 else:
12     print(f"{integer} is neither positive nor negative, it is zero.")
13
14 if integer % 2 == 0:
15     print(f"{integer} is an even number.")
16 else:
17     print(f"{integer} is an odd number.")
18

Problems  Output  Debug Console  Terminal  Ports
● PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB_1_TASKS\Lab_1_Task_1.py"
Enter a number: 24
24 is a positive number.
24 is an even number.
● PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB_1_TASKS\Lab_1_Task_1.py"
Enter a number: -99
-99 is a negative number.
-99 is an odd number.
❖ PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab>
```

2. Write a Python program that takes a number between 1–12 from the user and returns the month name. Use Python match-case.

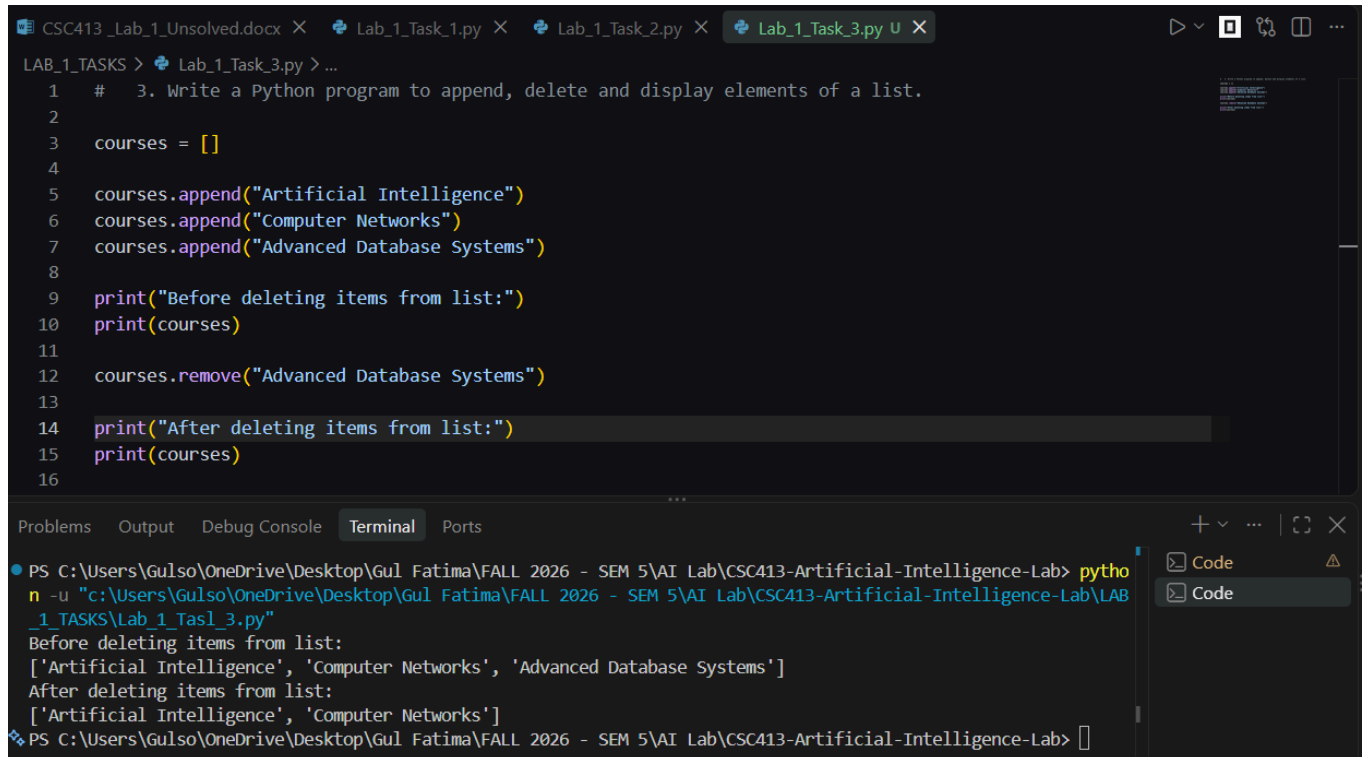
Code & Output:

```
CSC413_Lab_1_Unsolved.docx X Lab_1_Task_1.py X Lab_1_Task_2.py U X
LAB_1_TASKS > Lab_1_Task_2.py > ...
1 # 2. Write a Python program that takes a number between 1–12 from the user and returns the month name.
2 # Use Python match-case.
3
4 num = int(input("Enter a number between 1 to 12: "))
5
6 def check_number(num):
7     match num:
8         case 1:
9             return "January"
10        case 2:
11            return "February"
12        case 3:
13            return "March"
14        case 4:
15            return "April"
16        case 5:
17            return "May"
18        case 6:
19            return "June"
20        case 7:
21            return "July"
22        case 8:
23            return "August"
24        case 9:
25            return "September"
26        case 10:
27            return "October"
28        case 11:
29            return "November"
30        case 12:
31            return "December"
32        case _:
33            return "Invalid Month!"
34
35 print(check_number(num))
```

```
Problems Output Debug Console Terminal Ports
● PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python
n -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB
_1_TASKS\Lab_1_Task_2.py"
Enter a number between 1 to 12: 12
December
● PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python
n -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB
_1_TASKS\Lab_1_Task_2.py"
Enter a number between 1 to 12: 14
Invalid Month!
● PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python
n -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB
_1_TASKS\Lab_1_Task_2.py"
Enter a number between 1 to 12: 11
November
```

3. Write a Python program to append, delete and display elements of a list.

Code & Output:



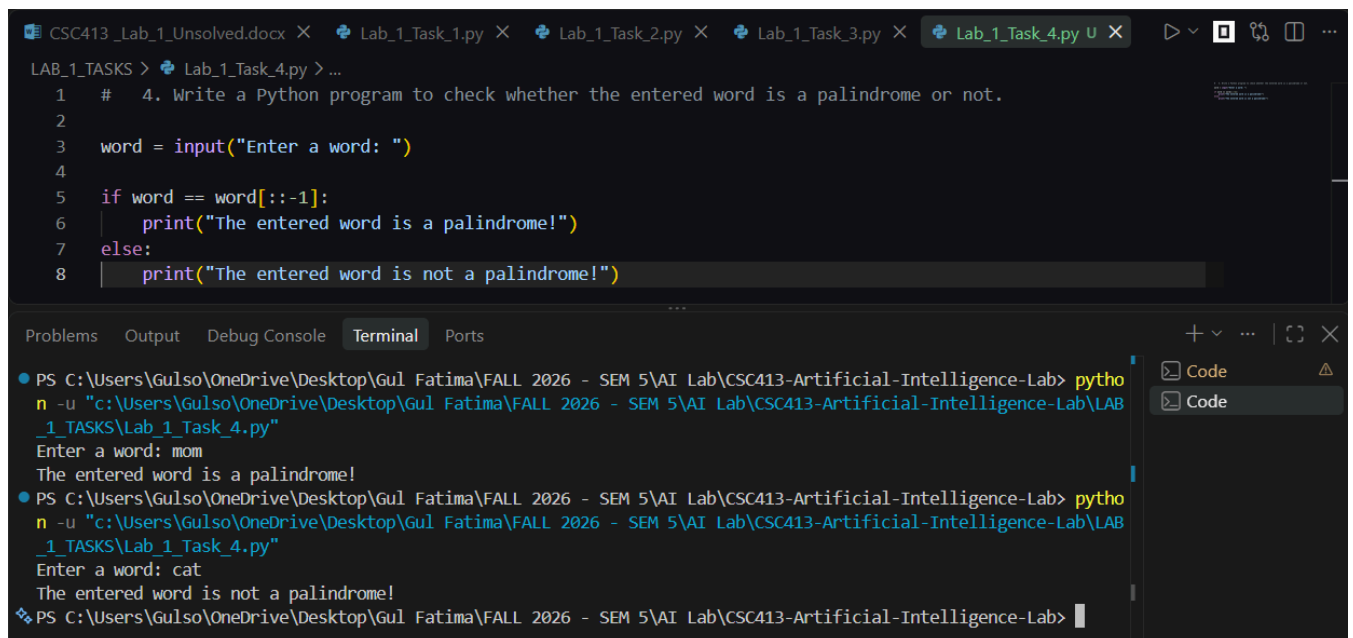
```
CSC413_Lab_1_Unsolved.docx X Lab_1_Task_1.py X Lab_1_Task_2.py X Lab_1_Task_3.py U X
LAB_1_TASKS > Lab_1_Task_3.py > ...
1 # 3. Write a Python program to append, delete and display elements of a list.
2
3 courses = []
4
5 courses.append("Artificial Intelligence")
6 courses.append("Computer Networks")
7 courses.append("Advanced Database Systems")
8
9 print("Before deleting items from list:")
10 print(courses)
11
12 courses.remove("Advanced Database Systems")
13
14 print("After deleting items from list:")
15 print(courses)
16
```

Terminal Output:

```
PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB_1_TASKS\Lab_1_Task_3.py"
Before deleting items from list:
['Artificial Intelligence', 'Computer Networks', 'Advanced Database Systems']
After deleting items from list:
['Artificial Intelligence', 'Computer Networks']
PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab>
```

4. Write a Python program to check whether the entered word is a palindrome or not.

Code & Output:



```
CSC413_Lab_1_Unsolved.docx X Lab_1_Task_1.py X Lab_1_Task_2.py X Lab_1_Task_3.py X Lab_1_Task_4.py U X
LAB_1_TASKS > Lab_1_Task_4.py > ...
1 # 4. Write a Python program to check whether the entered word is a palindrome or not.
2
3 word = input("Enter a word: ")
4
5 if word == word[::-1]:
6     print("The entered word is a palindrome!")
7 else:
8     print("The entered word is not a palindrome!")
```

Terminal Output:

```
PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB_1_TASKS\Lab_1_Task_4.py"
Enter a word: mom
The entered word is a palindrome!
PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab> python -u "c:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab\LAB_1_TASKS\Lab_1_Task_4.py"
Enter a word: cat
The entered word is not a palindrome!
PS C:\Users\Gulso\OneDrive\Desktop\Gul Fatima\FALL 2026 - SEM 5\AI Lab\CSC413-Artificial-Intelligence-Lab>
```

Conclusion:

In this lab, I practiced basic Python programming concepts through four different tasks. I used conditional statements to check whether a number is positive, negative, zero, even, or odd, and used match-case to return a month based on the given number. I also practiced working with lists by appending, deleting, and displaying elements, and used string comparison and slicing to check whether a word is a palindrome. Overall, these tasks helped me understand how different Python concepts can be combined to solve simple programming problems.

Student Deliverables

Submit the Python source code (.py or notebook), sample input/output or screenshots in a word file, and a brief conclusion describing the result and any observations.

Instructor/Assessment Focus

Correct Python syntax, logic, input/output, modularity, data structures and error handling.